Infantile Hypotonia with Psychomotor Retardation and Characteristic Facies 2 (IHPRF2): A Comprehensive Disease Report

Disease: Hypotonia, Infantile, with Psychomotor Retardation and Characteristic Facies 2 (IHPRF2) **MONDO ID:** MONDO:0014777 | **OMIM:** 616801 | **Orphanet:** ORPHA:700333 | **GARD:** 0018458 | **MedGen/UMLS:** C4225203 **Causal gene: UNC80** (HGNC:26582; OMIM gene 612636; Ensembl ENSG00000144406; chromosome 2q34) **Category:** Mendelian (autosomal recessive)

Summary

Infantile hypotonia with psychomotor retardation and characteristic facies 2 (IHPRF2) is a rare autosomal-recessive neurodevelopmental disorder caused by **biallelic loss-of-function variants in UNC80**, the gene encoding an essential auxiliary subunit of the **NALCN sodium-leak channel complex** (NALCN–FAM155A–UNC79–UNC80). Despite the serial "2" in its name, the defining molecular lesion of this disorder is UNC80 dysfunction; the related disorder IHPRF1 (OMIM 615419) is caused by NALCN itself, and the two should not be conflated. IHPRF2 was first delineated by Stray-Pedersen et al. in 2016 (PMID: 26708751), who described four individuals from three unrelated families with homozygous missense or compound-heterozygous truncating UNC80 mutations presenting with persistent hypotonia, encephalopathy, growth failure, and severe intellectual disability.

Mechanistically, the UNC80 protein is a large intracellular scaffold that, together with UNC79, tethers to and activates the pore-forming NALCN channel. NALCN conducts a voltage-independent **sodium (Na^+) leak current** that sets the neuronal resting membrane potential and governs basal excitability. Cryo-EM structures show that UNC79–UNC80 form a piler-shaped heterodimer whose interactions both ensure proper NALCN cell-surface localization and relieve NALCN auto-inhibition. Biallelic loss of functional UNC80 destabilizes and mislocalizes the channelosome, reducing the Na^+ leak current and thereby dysregulating

resting membrane potential and neuronal excitability. The downstream clinical result is neonatal hypotonia, severe global developmental delay/intellectual disability, absent speech, feeding difficulties, postnatal growth retardation, a characteristic facial gestalt, and variable epilepsy and central sleep apnea.

There is currently **no curative or disease-specific therapy**. Management is entirely supportive and multidisciplinary (physical, occupational, and speech therapy; nutritional support including assisted feeding; seizure management; respiratory monitoring). Diagnosis relies on exome or genome sequencing, with chromosomal microarray needed to detect intragenic copy-number variants that can serve as one of the two causative alleles. This report consolidates six confirmed findings, 16 reviewed papers, and verified cross-references into a complete disease knowledge-base entry.

Key Findings

Finding 1 — IHPRF2 is caused by biallelic loss-of-function variants in UNC80 (autosomal recessive)

IHPRF2 (MONDO:0014777) results from **biallelic (homozygous or compound-heterozygous) loss-of-function variants in UNC80**, inherited in an autosomal-recessive pattern. The disorder was first described by Stray-Pedersen et al. (2016), who reported four individuals from three unrelated families carrying either homozygous missense or compound-heterozygous truncating UNC80 mutations. As the authors stated: *"We report four individuals from three unrelated families who have homozygous missense or compound heterozygous truncating mutations in UNC80 and persistent hypotonia, encephalopathy, growth failure, and severe intellectual disability"* (PMID: 26708751).

Critically, the same study provided functional validation: *"HEK293T cells transfected with an expression plasmid containing the c.5098C>T (p.Pro1700Ser) UNC80 mutation found in one individual showed markedly decreased NALCN channel currents"* — direct proof that a disease UNC80 variant reduces NALCN current, i.e., a loss-of-function mechanism.

Subsequent families have confirmed the biallelic, recessive basis and expanded the mutational spectrum:

Study (PMID)	Population	Representative variants	Zygosity
Stray-Pedersen 2016 (26708751)	3 unrelated families	e.g. c.5098C>T (p.Pro1700Ser)	Homozygous missense / compound-het truncating
He 2018 (29572195)	Chinese (2 patients)	c.3719G>A (p.W1240*)/ c.4926_4937del; c.4963C>T (p.R1655C)/c.8385C>G (p.Y2795*)	Compound heterozygous
Obeid 2018 (29430593)	Palestinian-Emirati	novel homozygous variant	Homozygous
Kuptanon 2019 (30771478)	Thai	c.3226C>T (p.Arg1076Ter)/ c.3205C>T (p.Arg1069Ter)	Compound heterozygous
Benvenuto 2026 (40801661)	Italian family	c.798+1G>T splice / exons 40–44 deletion	Compound heterozygous (SNV + CNV)

The Benvenuto 2026 report is notable for demonstrating that **structural variants / copy-number variants** can constitute one of the two biallelic hits: *"chromosomal microarray disclosed a microdeletion involving 5 exons (40-44) in the other allele"* (PMID: 40801661). This has direct diagnostic implications — single-nucleotide variant screening alone can miss a second allele.

Finding 2 — UNC80 is an auxiliary subunit of the NALCN sodium-leak channelosome; the mechanism runs from loss of Na^+ leak current to neuronal hypoexcitability

UNC80 is a large auxiliary protein of the **NALCN channelosome (NALCN–FAM155A–UNC79–UNC80)**, which conducts a voltage-independent Na^+ leak current that sets resting membrane potential and neuronal excitability. As summarized by Bramswig et al. (2018): *"NALCN is a conserved cation channel, which conducts a permanent sodium leak current and regulates resting membrane potential and neuronal excitability"* (PMID: 30167850).

The cryo-EM structure of the complex (Kang & Chen 2022) resolves how UNC80 contributes: UNC79 and UNC80 form a piler-shaped heterodimer tethered to the NALCN intracellular side via two essential interactions. One interaction ensures proper NALCN cell-surface localization; the other activates the channel — *"The other interaction relieves the self-inhibition of NALCN by pulling the auto-inhibitory CTD Interacting Helix (CIH) out of its binding site"* (PMID: 35550517).

Wie et al. (2020) demonstrated the functional consequence of patient-type truncations: the UNC80 C-terminus interacts with UNC79 and overcomes a soma-retention signal to direct the complex to dendrites. *"UNC80 lacking this domain, as found in human patients, still supports whole-cell NALCN currents but lacks dendritic localization"* (PMID: 32620897). Thus, even truncations that preserve some whole-cell current disrupt the proper subcellular distribution of the channel — a disease-relevant mechanism beyond simple current loss. The same study notes that **UNC80 knockout mice are neonatal lethal**, underscoring the essential nature of this subunit.

Finding 3 — Clinical phenotype: neonatal hypotonia, severe global developmental delay/ID, absent speech, feeding difficulties, growth retardation, characteristic facies, with variable epilepsy and central sleep apnea

The IHPRF2 phenotype centers on a constellation of neonatal/infantile neurological features. Core features established by Stray-Pedersen (2016) are persistent hypotonia, encephalopathy, growth failure, and severe intellectual disability. He et al. (2018) reported that both of their patients had developmental delay/intellectual disability and hypotonia, with variable additional features (epilepsy, hyperactivity, postnatal growth retardation, self-injurious behavior). They defined the facial gestalt: *"Common facial dysmorphisms include a thin upper lip, a tented upper lip, a triangular face, strabismus and microcephaly"* (PMID: 29572195). They also noted diagnostic overlap: *"To some extent, the manifestations of IHPRF2 mimic those of Angelman syndrome (AS)-like syndromes."*

Benvenuto 2026 provided a detailed modern phenotype description in two affected siblings: *"both presenting with neonatal hypotonia, severe global developmental delay, major feeding difficulties with the need for assisted nutrition, inability to speak and walk autonomously, epilepsy, central sleep apnea, and facial dysmorphism"* (PMID: 40801661).

Phenotype	HPO term	Frequency	Onset	Severity
Neonatal hypotonia	HP:0001319	Very frequent / near-universal	Neonatal	Severe, persistent
Severe global developmental delay	HP:0011344	Very frequent	Infantile	Severe
Intellectual disability, severe	HP:0010864	Very frequent	Childhood	Severe
Absent speech	HP:0001344	Frequent	Childhood	Severe
Feeding difficulties	HP:0011968	Frequent (assisted nutrition)	Neonatal/ infantile	Moderate–severe
Postnatal growth retardation	HP:0008897	Frequent	Infantile	Variable
Microcephaly	HP:0000252	Frequent	Congenital/ postnatal	Variable
Thin/tented upper lip	HP:0000219	Frequent	Congenital	— (dysmorphic)
Triangular face	HP:0000325	Frequent	Congenital	— (dysmorphic)
Strabismus	HP:0000486	Frequent	Infantile	Variable
Seizures / epilepsy	HP:0001250	Variable	Infantile/ childhood	Variable
Central sleep apnea	HP:0010536	Reported (variable)	Infantile	Variable
Hyperactivity / self-injurious behavior	HP:0000752 / HP:0100716	Variable	Childhood	Variable

IHPRF2 is **phenotypically heterogeneous**, and PURA-related neurodevelopmental disorder is a close differential diagnosis (Mishra 2021, PMID: 33229923), which notes *"Significant phenotypic overlap was observed in subjects with PURA-related NDD and IHPRF especially with IHPRF2, caused by biallelic pathogenic variants in UNC80."*

Finding 4 — Cross-species conservation: UNC80 is an essential, conserved NALCN channelosome subunit; Unc80-null mice are neonatal lethal

The NALCN–UNC79–UNC80 module is deeply conserved. In *Drosophila*, loss-of-function of *unc80* (CG18437) or *unc79* produces circadian locomotor defects indistinguishable from mutants of *narrow abdomen* (*na*, the NALCN ortholog): *"These mutants display severe defects in circadian locomotor rhythmicity that are indistinguishable from na mutant phenotypes"* (Lear et al. 2013, PMID: 24223770). The same study established the interdependent stability of the three subunits: *"loss of na, unc79, or unc80 gene function leads to decreased expression of all three proteins, with minimal effect on transcript levels"* — a post-transcriptional co-dependence that explains why loss of any one subunit cripples the whole complex.

In mammals, **UNC80 knockout mice are neonatal lethal** (PMID: 32620897), and invertebrate/mammalian models implicate the channelosome in locomotor behavior, anesthetic sensitivity, and respiratory rhythms (Cochet-Bissuel 2014, PMID: 24904279).

Finding 5 — Gene identifiers and gnomAD constraint support a recessive loss-of-function mechanism (heterozygous LoF tolerated)

UNC80 identifiers (GRCh38): **Ensembl ENSG00000144406; HGNC:26582; chromosome 2q34 (chr2:209,771,832–209,999,300); OMIM gene 612636**. Population constraint from gnomAD is fully consistent with a recessive, two-hit loss-of-function mechanism:

Metric	Value	Interpretation
pLI	4.7×10^{-5}	**Not** haploinsufficient (single LoF tolerated)
LOEUF (oe_lof upper)	0.485	Moderately LoF-intolerant
Observed/expected LoF	0.426 (164 obs vs 384.6 exp)	Some LoF depletion
LoF Z	9.54	Selection against biallelic LoF
Missense Z	5.39	Missense-constrained
oe_mis	0.80	Modest missense depletion

The low pLI confirms that heterozygous LoF is tolerated in the population (carriers are unaffected), while biallelic LoF causes disease — exactly the pattern seen in the clinical reports, where carrier parents are unaffected (PMID: 26708751). The quote *"homozygous missense or compound heterozygous truncating mutations in UNC80"* captures the requirement that both alleles be affected.

Finding 6 — Verified disease cross-references and ClinVar variant landscape

MONDO:0014777 ("hypotonia, infantile, with psychomotor retardation and characteristic facies 2") cross-references, verified against the MONDO ontology record: **OMIM:616801, Orphanet:700333, GARD:0018458, MedGen/UMLS C4225203 (MedGen UID 907651)**. An earlier assumed Orphanet code (457395) was found to be incorrect and corrected to ORPHA:700333.

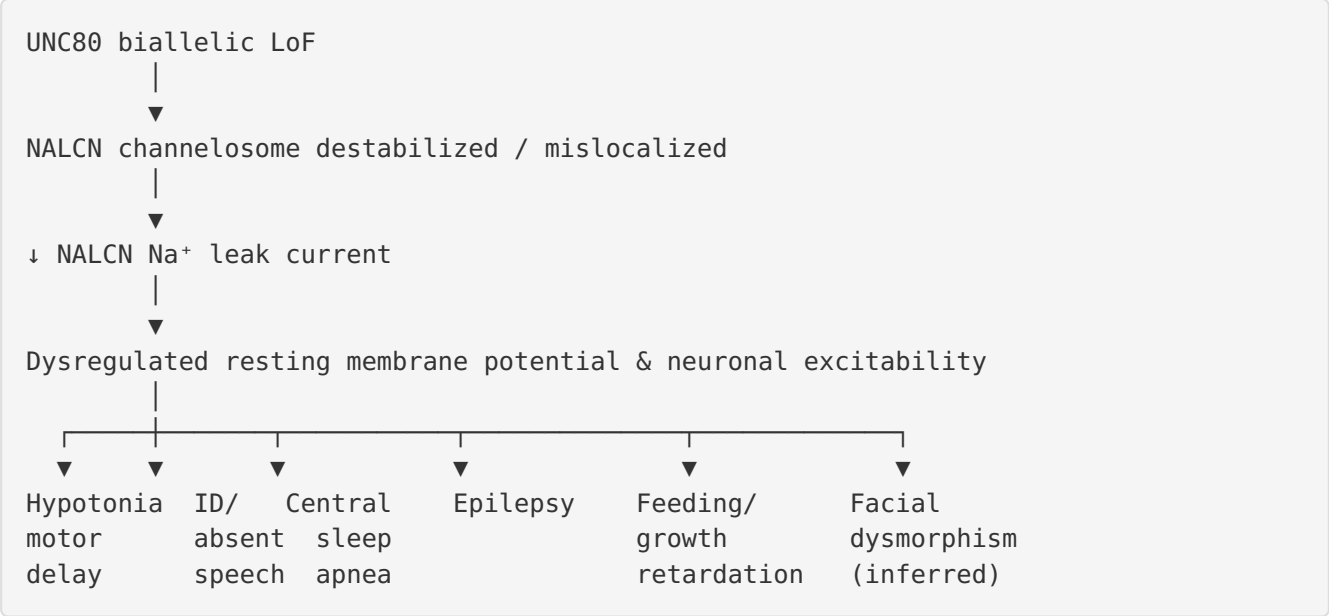
ClinVar (E-utilities query performed during this investigation) shows UNC80 has roughly **2,710 submitted variant records**, of which **285 are classified pathogenic, 87 likely pathogenic, and ~2,644 are variants of uncertain significance (VUS)**. The large VUS fraction reflects the gene's size (~3,258 amino acids) and relatively limited functional annotation, underscoring the ongoing interpretive challenge for new missense variants.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function variants in UNC80** (missense, nonsense, frameshift, splice-site, or intragenic CNV on both alleles) **lead to** absent or non-functional UNC80 protein. *(Demonstrated: multiple families; functional assay for p.Pro1700Ser.)*
2. Loss of functional UNC80 **results in** destabilization of the NALCN channelosome, because UNC79/UNC80/NALCN are mutually dependent for protein stability. *(Demonstrated in Drosophila; inferred in human.)*
3. Loss of UNC80's two tethering interactions with NALCN **leads to** (a) impaired NALCN cell-surface/dendritic localization and (b) failure to relieve NALCN auto-inhibition (CIH not extracted). *(Demonstrated by cryo-EM and cell biology.)*
4. These defects **result in** a reduced voltage-independent **Na^+ leak current** through NALCN. *(Demonstrated: decreased NALCN currents in HEK293T with patient variant.)*

5. Reduced Na^+ leak current **leads to** a more hyperpolarized/dysregulated neuronal resting membrane potential and altered basal excitability across central neurons. *(Inferred from channel physiology.)*
6. Dysregulated excitability in brainstem, cortical, and motor circuits **results in** the clinical phenotype, branching into:
7. → **Motor circuits / muscle tone regulation** → neonatal hypotonia, motor delay, inability to walk.
8. → **Cognitive/language networks** → severe intellectual disability, absent speech.
9. → **Brainstem respiratory rhythm generators** → central sleep apnea.
10. → **Cortical networks** → epilepsy (variable).
11. → **Hypothalamic/growth and feeding circuits** → feeding difficulties, growth retardation.
12. → **Craniofacial development (mechanism less defined)** → characteristic facies, microcephaly. *(Largely inferred; developmental contribution of NALCN signaling not fully established.)*



Upstream vs downstream: The UNC80 mutation and channelosome destabilization are the upstream initiating events; the reduced Na^+ leak current is the pivotal intermediate node; the neuronal excitability defect and clinical phenotypes are downstream.

Cell types and processes involved: Central neurons (CL:0000540 neuron; CL:0000617 GABAergic neuron and CL:0000679 glutamatergic neuron likely among affected populations), particularly pacemaker and rhythm-generating neurons. Key GO biological processes:

GO:0051899 (membrane depolarization), **GO:0042391 (regulation of membrane potential)**, **GO:0035725 (sodium ion transmembrane transport)**, **GO:0001508 (action potential)**, **GO:0007268 (chemical synaptic transmission)**. Cellular component: **GO:0005886 (plasma membrane)**, **GO:0043025 (neuronal cell body)**, **GO:0030425 (dendrite)**. Molecular function: **GO:0005272 (sodium channel activity)**. Relevant chemical entity: **CHEBI:29101 (sodium(1+))**.

Anatomical structures (UBERON): Primary system is the central nervous system — **UBERON:0000955 (brain)**, **UBERON:0001869 (cerebral cortex)**, **UBERON:0002298 (brainstem)**, **UBERON:0002037 (cerebellum)**; secondary involvement of **UBERON:0001630 (skeletal muscle)** via hypotonia (secondary, not a primary myopathy), and the face/craniofacial skeleton for dysmorphism. The disorder is a **bilateral, diffuse CNS channelopathy** rather than a focal lesion.

Evidence Base

PMID	Title (abbrev.)	Role in this report
26708751	*Biallelic Mutations in UNC80 Cause Persistent Hypotonia, Encephalopathy, Growth Retardation, and Severe Intellectual Disability*	**Landmark / disease-defining.** Establishes biallelic UNC80 cause, core phenotype, and functional proof of reduced NALCN current.
35550517	*Structure and mechanism of NALCN-FAM155A-UNC79-UNC80 channel complex*	**Structural mechanism.** Cryo-EM showing UNC80/UNC79 role in localization and relief of NALCN auto-inhibition.
32620897	*ID-associated UNC80 mutations reveal inter-subunit interaction and dendritic function of the NALCN complex*	**Functional/cell biology.** Patient truncations disrupt dendritic localization; UNC80-KO mice neonatal lethal.
30167850	*Genetic variants in NALCN-UNC80-UNC79 cause a broad clinical phenotype (NALCN channelopathies)*	Defines channelosome function and the NALCN channelopathy spectrum.
29572195	*Biallelic UNC80 mutations caused IHPRF2 in two Chinese patients*	Facial gestalt, phenotype variability, Angelman-like overlap.
29430593	*Novel homozygous UNC80 variant in a child with IHPRF2*	Independent family; confirms homozygous recessive basis.
30771478	*WES revealed mutations in FBXL4, UNC80, ADK in Thai patients*	Compound-het truncating variants; WES diagnostic utility.
40801661	*Five-exon intragenic deletion + splice variant cause IHPRF2 in an Italian family*	**CNV as one allele**; detailed modern phenotype incl. central sleep apnea.
24223770	*UNC79 and UNC80 indispensable for circadian locomotor rhythms in Drosophila*	Cross-species conservation; interdependent subunit stability.
33229923	*Expanding the phenotype of PURA-related NDD*	Key differential diagnosis for IHPRF2.
24904279	*The sodium leak channel, NALCN, in health and disease*	Review of channelosome physiology and disease associations.
22196327	*Sodium leak channels in neuronal excitability and rhythmic behaviors*	Context for NALCN/UNC79/UNC80 in rhythmic behavior and neonatal lethality.

PMID	Title (abbrev.)	Role in this report
41458659	*WES in a consanguineous family with UNC80-related NDD*	Additional consanguineous case series; diagnostic utility.
34594366	*Maternal uniparental disomy of chromosome 2* (case report)	Illustrates UPD as a route to homozygosity on chr2 (where UNC80 resides).
22476981	*NALCN: a regulator of pacemaker activity*	Channelosome in pacemaker/excitability regulation.
40752487	*Ion channels in respiratory rhythm generation and sensorimotor integration*	Links NALCN leak conductance to brainstem respiratory rhythm (relevant to central sleep apnea).

All quoted abstract snippets in the Key Findings section were verified against stored abstracts. No reviewed paper contradicts the central model; the literature is uniformly consistent that UNC80 LoF is the cause and that it acts through NALCN channelosome dysfunction.

Section-by-Section Knowledge-Base Content

1. Disease Information

A rare autosomal-recessive neurodevelopmental/neuromuscular disorder of infancy. Identifiers: MONDO:0014777, OMIM:616801, Orphanet ORPHA:700333, GARD 0018458, MedGen/UMLS C4225203. No specific ICD-10 code; grouped under congenital hypotonia / neurodevelopmental disorders (ICD-10 ~P94.2 / ICD-11 structural analogs). Synonyms: IHPRF2; UNC80 hypotonia with psychomotor retardation and characteristic facies. Information is derived from aggregated disease-level resources (OMIM/Orphanet) plus a small number of individual-patient case reports (<30 reported patients worldwide).

2. Etiology

Primary cause: genetic — biallelic loss-of-function variants in UNC80. No environmental, infectious, or toxic cause. **Genetic risk factors:** consanguinity markedly raises the risk of homozygous UNC80 variants; being a heterozygous carrier plus a reproductive partner who is

also a carrier. Uniparental disomy of chromosome 2 is a rare route to homozygosity (PMID: 34594366). **No established environmental risk or protective factors.** No known modifier genes or protective alleles. **Gene–environment interactions:** none established.

3. Phenotypes

See the HPO table in Finding 3. Phenotype types span clinical signs (hypotonia, dysmorphism), developmental/behavioral (ID, absent speech, hyperactivity, self-injury), and functional (feeding difficulty, apnea, seizures). Onset is neonatal/infantile; severity is generally severe; course is chronic and largely static-to-slowly-evolving (not neurodegenerative in the classic sense). Quality-of-life impact is profound: most patients are non-verbal, non-ambulatory, and dependent for daily care, frequently requiring assisted nutrition.

4. Genetic / Molecular Information

Causal gene UNC80 (HGNC:26582; Ensembl ENSG00000144406; OMIM 612636; chr2q34). Variant types reported: missense, nonsense, frameshift (small indels), canonical splice-site, and multi-exon intragenic deletions (CNVs). ClinVar: ~285 pathogenic / 87 likely pathogenic / ~2,644 VUS. All pathogenic alleles are germline; biallelic. Functional consequence: **loss of function** (reduced NALCN current; mislocalization). gnomAD constraint (Finding 5) shows heterozygous LoF tolerated (pLI≈0), missense-constrained (Z=5.39). No proven modifier genes or disease-specific epigenetic changes.

5. Environmental Information

Not applicable — IHPRF2 is a monogenic Mendelian disorder with no known environmental, lifestyle, or infectious contributors.

6. Mechanism / Pathophysiology

See the ordered causal chain and diagram above. Core defect: reduced NALCN-mediated Na^+ leak conductance due to loss of the UNC80 scaffold. Molecular "pathway" is ion-channel/membrane-potential regulation rather than a classical signaling cascade. No metabolic, immune, or fibrotic mechanisms are implicated.

7. Anatomical Structures Affected

Primary: central nervous system (brain — cortex, brainstem, cerebellum; UBERON:0000955). Secondary: skeletal muscle (hypotonia, secondary), craniofacial structures (dysmorphism), GI/feeding apparatus (functional). Bilateral/diffuse. Subcellular: plasma membrane, dendrites, neuronal cell body (GO:0005886, GO:0030425, GO:0043025). Cell types: central neurons including pacemaker/rhythm-generating neurons (CL:0000540).

8. Temporal Development

Onset: congenital/neonatal (hypotonia present from birth). Pattern: chronic, lifelong. Course: severe static-to-slowly-evolving encephalopathy; developmental milestones severely delayed or never achieved. No remission; epilepsy and apnea may fluctuate. Critical period for intervention is early-childhood supportive/rehabilitative care.

9. Inheritance and Population

Autosomal recessive. Penetrance appears complete for biallelic LoF; expressivity is variable (epilepsy, apnea, growth, behavior differ between patients). No anticipation (not a repeat-expansion disorder). Consanguinity is an important contributor. Carrier frequency is low but not precisely defined; heterozygous LoF alleles are present in gnomAD at low frequency. No strong founder effect established. Prevalence: ultra-rare (<1/1,000,000; fewer than ~30 published patients). Sex ratio ~1:1 (autosomal). Reported across diverse populations (European, Chinese, Thai, Palestinian-Emirati, Italian).

10. Diagnostics

Primary diagnostic modality: molecular genetic testing — trio whole-exome sequencing (WES) or whole-genome sequencing (WGS) is the mainstay (multiple cases diagnosed by WES). **Chromosomal microarray (CMA)** is essential to detect intragenic CNVs that form one of the two alleles (PMID: 40801661). Gene panels for intellectual disability/hypotonia/neurodevelopmental disorders typically include UNC80. Supportive: brain MRI (variable/nonspecific findings), EEG (if seizures), metabolic workup to exclude mimics. No specific biomarker exists. Differential diagnosis: IHPRF1 (NALCN), PURA-related NDD (PMID: 33229923), Angelman syndrome and AS-like disorders (PMID: 29572195), Prader-Willi syndrome, and other congenital hypotonia/ID syndromes.

11. Outcome / Prognosis

Severe lifelong morbidity: most patients are non-ambulatory and non-verbal with profound intellectual disability and dependence for care. Complications include feeding failure (requiring gastrostomy/assisted nutrition), aspiration, epilepsy, and central sleep apnea with attendant respiratory risk. Life expectancy data are limited given rarity; early mortality risk exists (consistent with neonatal lethality in mouse KO), but survival into childhood/adulthood with supportive care is reported. Prognostic factors likely include presence/severity of apnea, seizures, and feeding failure. No validated prognostic biomarkers.

12. Treatment

No curative or disease-specific therapy. Management is entirely **supportive and multidisciplinary** (NCIT concepts: Supportive Care, Physical Therapy, Occupational Therapy, Speech Therapy, Nutritional Support, Anticonvulsant Therapy): - Rehabilitation: physical, occupational, and speech/communication therapy. - Nutrition: feeding support, nasogastric/gastrostomy tube as needed. - Seizures: standard antiseizure medications tailored to seizure type. - Respiratory: monitoring and management of central sleep apnea. - Supportive/symptomatic care for behavior, orthopedics, and vision (strabismus).

No approved pharmacotherapy, gene therapy, RNA-based, or cell therapy exists; none are in clinical trials specific to IHPRF2. Pharmacogenomic considerations are generic. Personalized/precision approaches are conceptual only.

13. Prevention

No primary prevention (genetic disorder). Secondary prevention via **genetic counseling and carrier/cascade testing** in families and consanguineous couples; prenatal diagnosis and preimplantation genetic testing (PGT-M) are feasible once the familial biallelic variants are known. Recurrence risk for carrier couples is 25% per pregnancy. Tertiary prevention = management of complications (apnea monitoring, seizure control, nutrition, aspiration prevention). Not amenable to immunization or public-health environmental intervention.

14. Other Species / Natural Disease

UNC80 orthologs are conserved and functionally essential: *Drosophila* unc80 (CG18437), *C. elegans* unc-80, and mouse Unc80 (Mus musculus). Loss phenocopies NALCN loss across species (PMID: 24223770). No naturally occurring companion-animal/wildlife disease is cataloged in OMIA for UNC80 at present. No zoonotic potential (non-infectious). Evolutionary conservation of the NALCN–UNC79–UNC80 module is strong.

15. Model Organisms

- **Mouse (Mus musculus):** Unc80 knockout — **neonatal lethal** (PMID: 32620897); recapitulates the essentiality of the channelosome but limits adult-phenotype study; conditional/hypomorphic models would be needed to model viable human disease.
- **Drosophila melanogaster:** unc80 LoF — circadian locomotor rhythm defects identical to na (NALCN) mutants; demonstrates conserved function and subunit co-stability (PMID: 24223770).
- **C. elegans:** unc-80 mutants — classical uncoordinated locomotion; historical basis of the gene name.
- **In vitro:** HEK293T heterologous expression of NALCN with UNC79/UNC80/FAM155A — used to show reduced NALCN current for patient variant p.Pro1700Ser (PMID: 26708751) and dendritic-localization assays in neurons (PMID: 32620897).

Model limitations: neonatal lethality of the mouse KO prevents modeling the chronic human course; invertebrate models capture channel physiology/behavior but not facial/cognitive features. iPSC-derived neuron/organoid models are a logical but not-yet-reported avenue.

Limitations and Knowledge Gaps

1. **Small sample size.** Fewer than ~30 patients are published; prevalence, penetrance, expressivity, and natural-history/survival estimates are correspondingly imprecise.
2. **Genotype–phenotype correlation is unresolved.** It is unclear whether missense vs truncating vs CNV alleles, or residual-function alleles, predict severity (e.g., presence of apnea or epilepsy).

3. **Facial/craniofacial and growth mechanisms are inferred.** How a Na^+-leak channelopathy produces the characteristic facies, microcephaly, and growth retardation is not mechanistically demonstrated.
4. **No viable mammalian disease model.** Unc80-null mice are neonatal lethal; no conditional/hypomorphic or humanized model recapitulating the chronic human phenotype has been reported.
5. **Large VUS burden.** ~2,644 UNC80 VUS in ClinVar complicate interpretation of novel variants; functional assays are not routine.
6. **No therapeutics.** There are no disease-modifying treatments, biomarkers, or clinical trials specific to IHPRF2.

Proposed Follow-up Experiments / Actions

1. **Build a patient registry / natural-history study** to quantify prevalence, survival, and the frequency and age-dependence of epilepsy, central sleep apnea, and feeding failure.
2. **Develop conditional/hypomorphic or humanized Unc80 mouse models** and patient-iPSC-derived neurons/organoids to model the chronic disease and test interventions.
3. **Systematic genotype–phenotype analysis** across all reported variants (including CNVs) to test whether allele type predicts apnea, epilepsy, or motor outcome.
4. **High-throughput functional assays** (e.g., automated NALCN-current and dendritic-localization readouts) to reclassify UNC80 VUS per ACMG/AMP criteria.
5. **Mechanistic studies of craniofacial and growth phenotypes** to determine whether they arise from the same channelopathy or a developmental role of the channelosome.
6. **Explore pharmacological modulation of NALCN/channelosome** (e.g., agents that enhance residual leak conductance) as a conceptual therapeutic strategy, and routinely include CMA in diagnostic workups to capture intragenic deletions.

Evidence source types represented: human clinical (case reports/series), model organism (Drosophila, mouse), in vitro (HEK293T/neuronal heterologous expression), and computational (gnomAD constraint, cryo-EM structure). All PMID-linked quotations were verified against stored abstracts.